EDA Using Power BI

Project Title: Supply Chain & Inventory

Problem Statement :

A manufacturing company manages inventory across multiple warehouses. The company faces two major issues:

- Stock-outs causing delayed customer deliveries
- Overstocking leading to high holding costs

Currently, inventory decisions are made manually without analytical support.
As a Supply Chain Analyst, you are asked to:

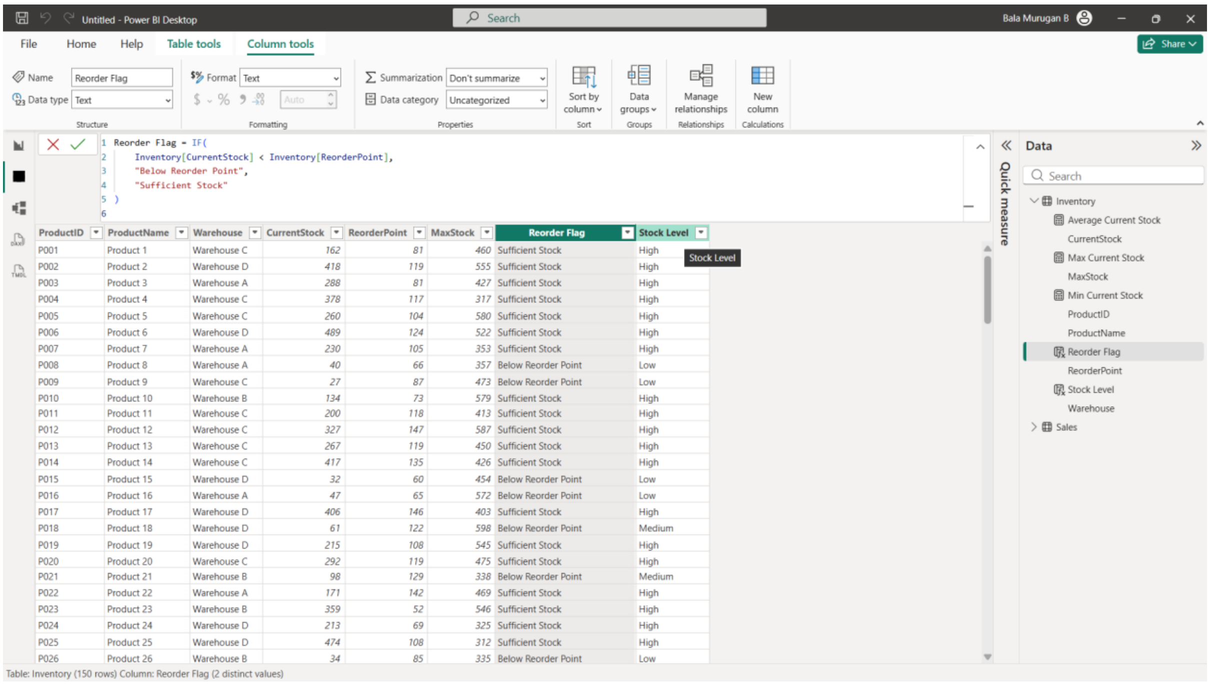
- Monitor stock levels
- Identify low-stock and overstocked products
- Optimize reorder decisions
- Improve inventory turnover

Questions:

1. Identify products with stock levels below reorder point.
2. Perform univariate analysis on stock levels.
3. Which warehouses experience frequent stock shortages?
4. Create a DAX measure to classify stock status (Low, Optimal, Excess).
5. Calculate inventory turnover rate.
6. Identify slow-moving products using sales and inventory data.
7. Design a dashboard to monitor stock health.
8. Which products require immediate replenishment?
9. How can inventory insights reduce holding costs?

1.Identify products with stock levels below reorder point:

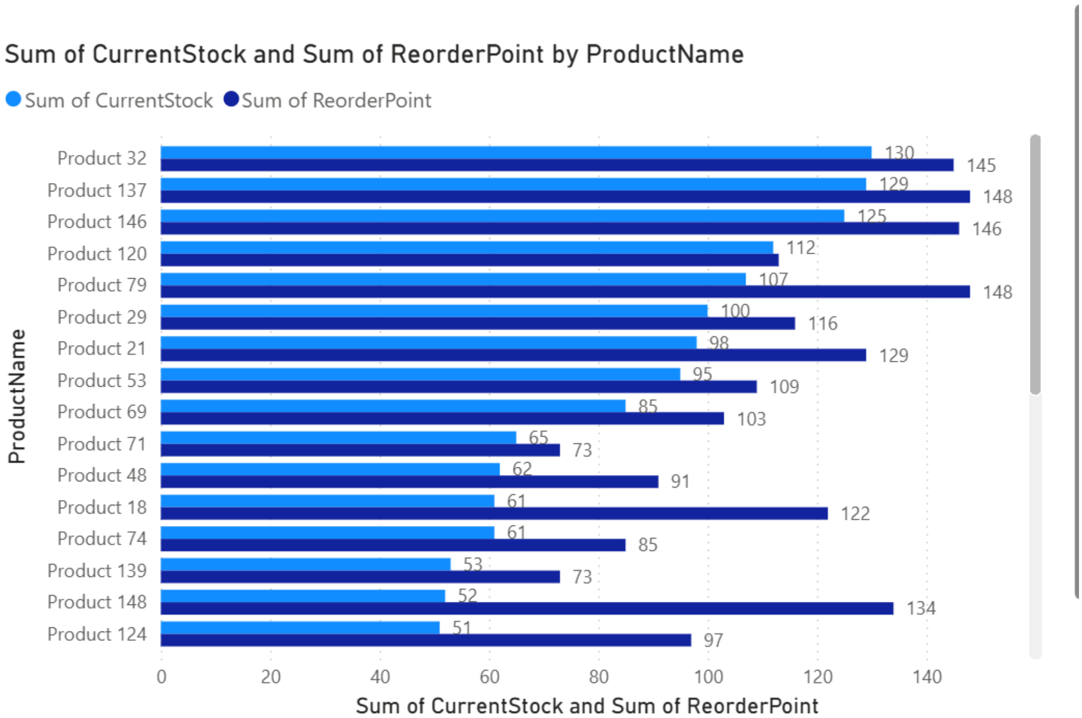
This analysis helps detect products that are running low in inventory and may soon run out, ensuring timely replenishment and avoiding stockouts.



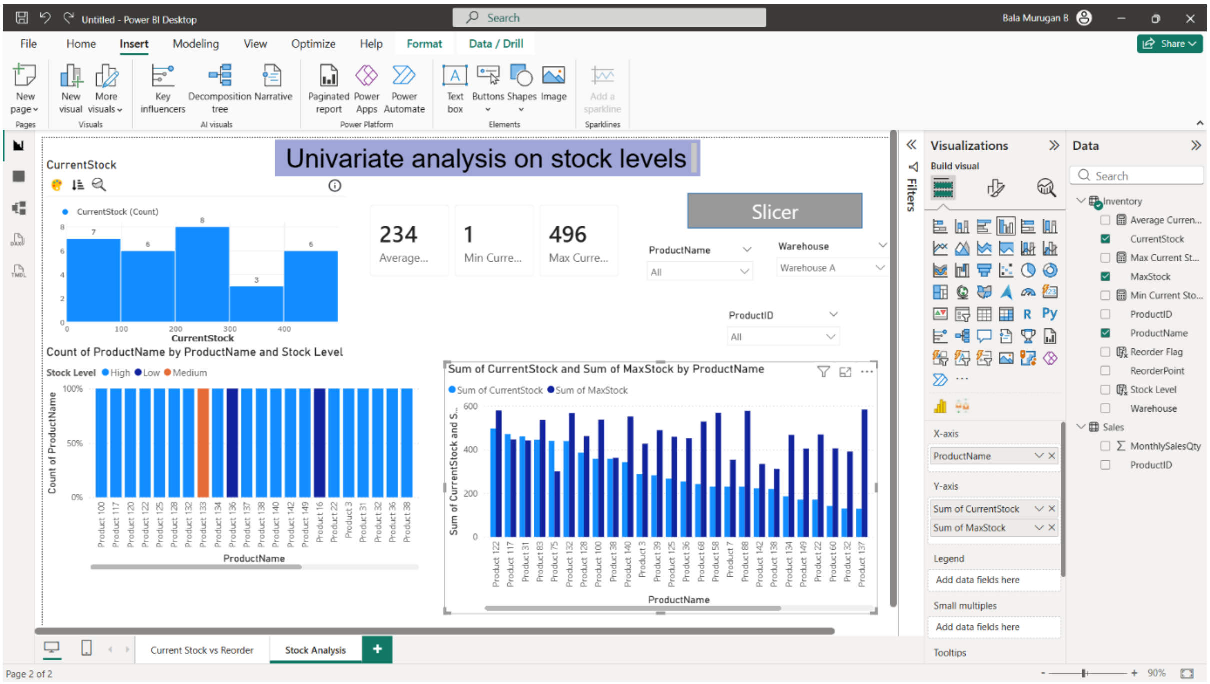
```
Reorder Flag = IF(
    Inventory[CurrentStock] < Inventory[ReorderPoint],
    "Below Reorder Point",
    "Sufficient Stock"
)
```

ProductID	ProductName	Warehouse	CurrentStock	ReorderPoint	MaxStock	Reorder Flag	Stock Level
P001	Product 1	Warehouse C	162	81	460	Sufficient Stock	High
P002	Product 2	Warehouse D	418	119	555	Sufficient Stock	High
P003	Product 3	Warehouse A	288	81	427	Sufficient Stock	High
P004	Product 4	Warehouse C	378	117	317	Sufficient Stock	High
P005	Product 5	Warehouse C	260	104	580	Sufficient Stock	High
P006	Product 6	Warehouse D	489	124	522	Sufficient Stock	High
P007	Product 7	Warehouse A	230	105	353	Sufficient Stock	High
P008	Product 8	Warehouse A	40	66	357	Below Reorder Point	Low
P009	Product 9	Warehouse C	27	87	473	Below Reorder Point	Low
P010	Product 10	Warehouse B	134	73	579	Sufficient Stock	High
P011	Product 11	Warehouse C	200	118	413	Sufficient Stock	High
P012	Product 12	Warehouse C	327	147	587	Sufficient Stock	High
P013	Product 13	Warehouse C	267	119	450	Sufficient Stock	High
P014	Product 14	Warehouse C	417	135	426	Sufficient Stock	High
P015	Product 15	Warehouse D	32	60	454	Below Reorder Point	Low
P016	Product 16	Warehouse A	47	65	572	Below Reorder Point	Low
P017	Product 17	Warehouse D	406	146	403	Sufficient Stock	High
P018	Product 18	Warehouse D	61	122	598	Below Reorder Point	Medium
P019	Product 19	Warehouse D	215	108	545	Sufficient Stock	High
P020	Product 20	Warehouse C	292	119	475	Sufficient Stock	High
P021	Product 21	Warehouse B	98	129	338	Below Reorder Point	Medium
P022	Product 22	Warehouse A	171	142	469	Sufficient Stock	High
P023	Product 23	Warehouse B	359	52	546	Sufficient Stock	High
P024	Product 24	Warehouse D	213	69	325	Sufficient Stock	High
P025	Product 25	Warehouse D	474	108	312	Sufficient Stock	High
P026	Product 26	Warehouse B	34	85	335	Below Reorder Point	Low

Table: Inventory (150 rows) Column: Reorder Flag (2 distinct values)

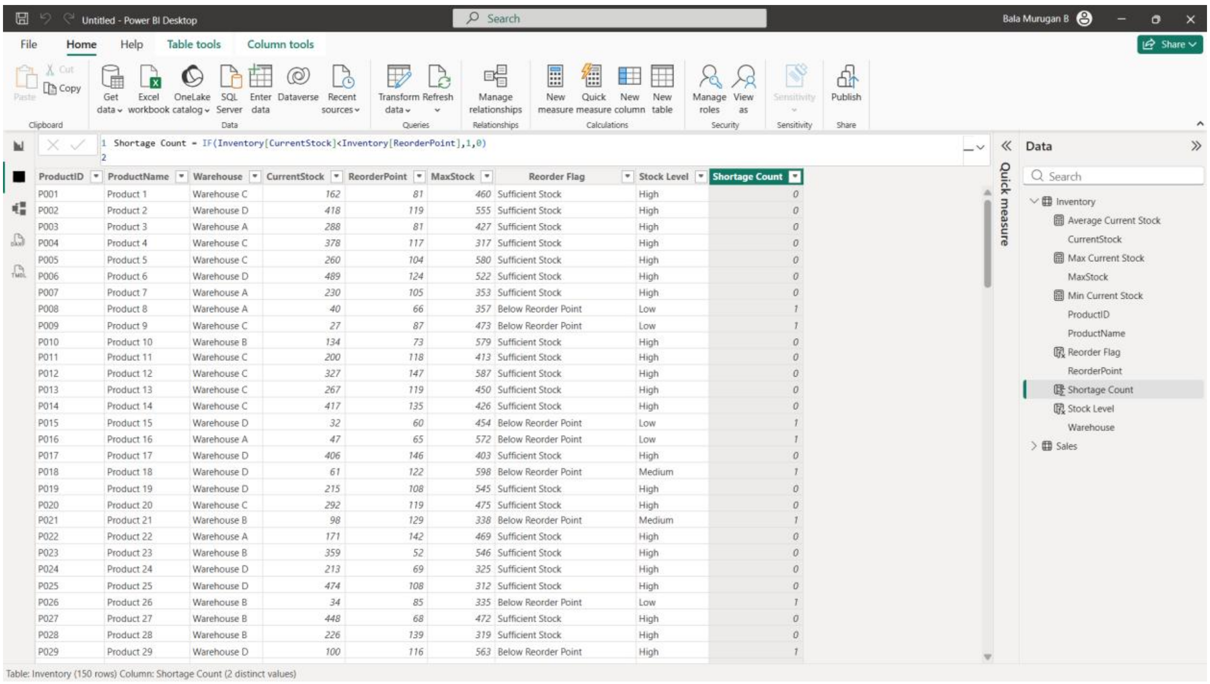


2.Perform univariate analysis on stock levels:



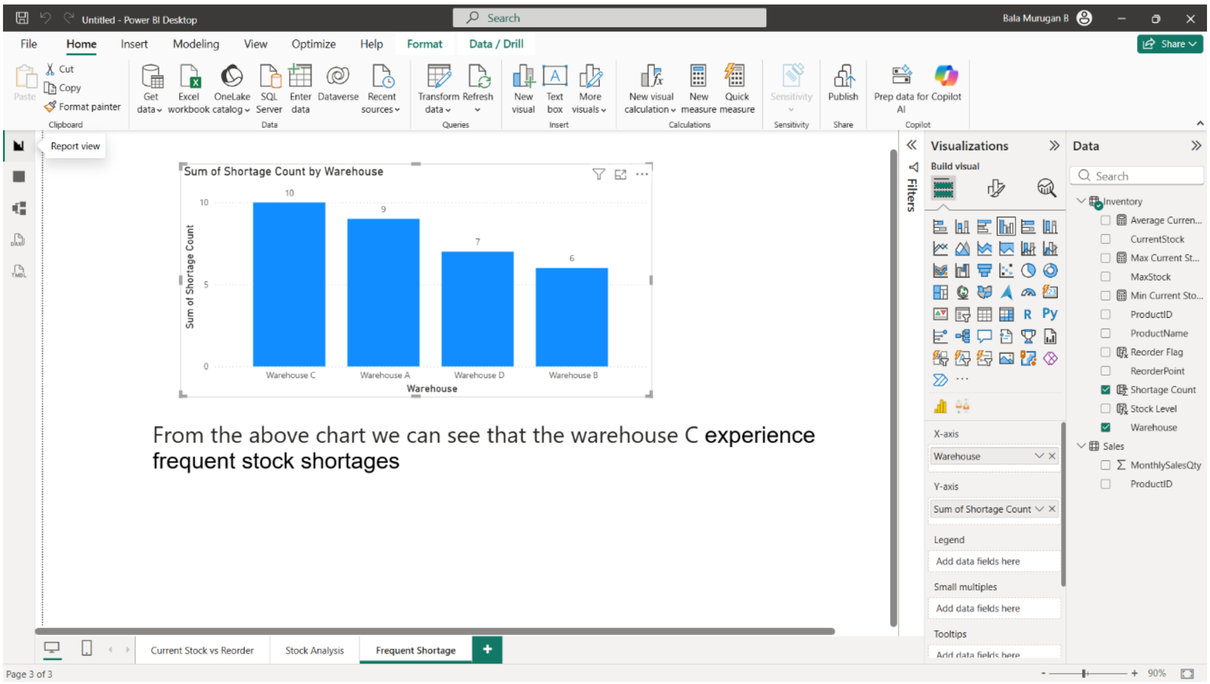
3.Which warehouses experience frequent stock shortages?

Frequent stock shortages at warehouses can lead to delayed deliveries, lost sales, and dissatisfied customers. Analyzing warehouse-level inventory data helps in targeted replenishment and better supply chain planning.



ProductID	ProductName	Warehouse	CurrentStock	ReorderPoint	MaxStock	Reorder Flag	Stock Level	Shortage Count
P001	Product 1	Warehouse C	162	81	460	Sufficient Stock	High	0
P002	Product 2	Warehouse D	418	119	555	Sufficient Stock	High	0
P003	Product 3	Warehouse A	288	81	427	Sufficient Stock	High	0
P004	Product 4	Warehouse C	378	117	317	Sufficient Stock	High	0
P005	Product 5	Warehouse C	260	104	580	Sufficient Stock	High	0
P006	Product 6	Warehouse D	489	124	522	Sufficient Stock	High	0
P007	Product 7	Warehouse A	230	105	353	Sufficient Stock	High	0
P008	Product 8	Warehouse A	40	66	357	Below Reorder Point	Low	1
P009	Product 9	Warehouse C	27	87	473	Below Reorder Point	Low	1
P010	Product 10	Warehouse B	134	73	579	Sufficient Stock	High	0
P011	Product 11	Warehouse C	200	118	413	Sufficient Stock	High	0
P012	Product 12	Warehouse C	327	147	587	Sufficient Stock	High	0
P013	Product 13	Warehouse C	267	119	450	Sufficient Stock	High	0
P014	Product 14	Warehouse C	417	135	426	Sufficient Stock	High	0
P015	Product 15	Warehouse D	32	60	454	Below Reorder Point	Low	1
P016	Product 16	Warehouse A	47	65	572	Below Reorder Point	Low	1
P017	Product 17	Warehouse D	406	146	403	Sufficient Stock	High	0
P018	Product 18	Warehouse D	61	122	598	Below Reorder Point	Medium	1
P019	Product 19	Warehouse D	215	108	545	Sufficient Stock	High	0
P020	Product 20	Warehouse C	292	119	475	Sufficient Stock	High	0
P021	Product 21	Warehouse B	98	129	338	Below Reorder Point	Medium	1
P022	Product 22	Warehouse A	171	142	469	Sufficient Stock	High	0
P023	Product 23	Warehouse B	359	52	546	Sufficient Stock	High	0
P024	Product 24	Warehouse D	213	69	325	Sufficient Stock	High	0
P025	Product 25	Warehouse D	474	108	312	Sufficient Stock	High	0
P026	Product 26	Warehouse B	34	85	335	Below Reorder Point	Low	1
P027	Product 27	Warehouse B	448	68	472	Sufficient Stock	High	0
P028	Product 28	Warehouse B	226	139	319	Sufficient Stock	High	0
P029	Product 29	Warehouse D	100	116	563	Below Reorder Point	High	1

Table: Inventory (150 rows) Column: Shortage Count (2 distinct values)

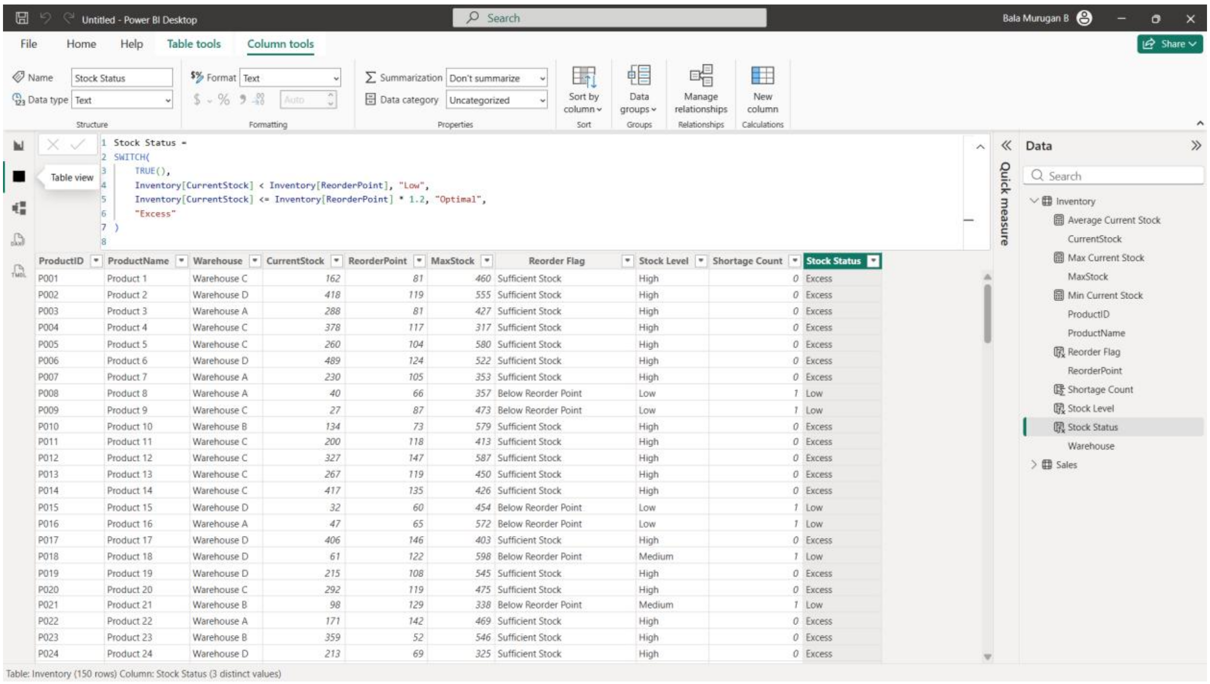


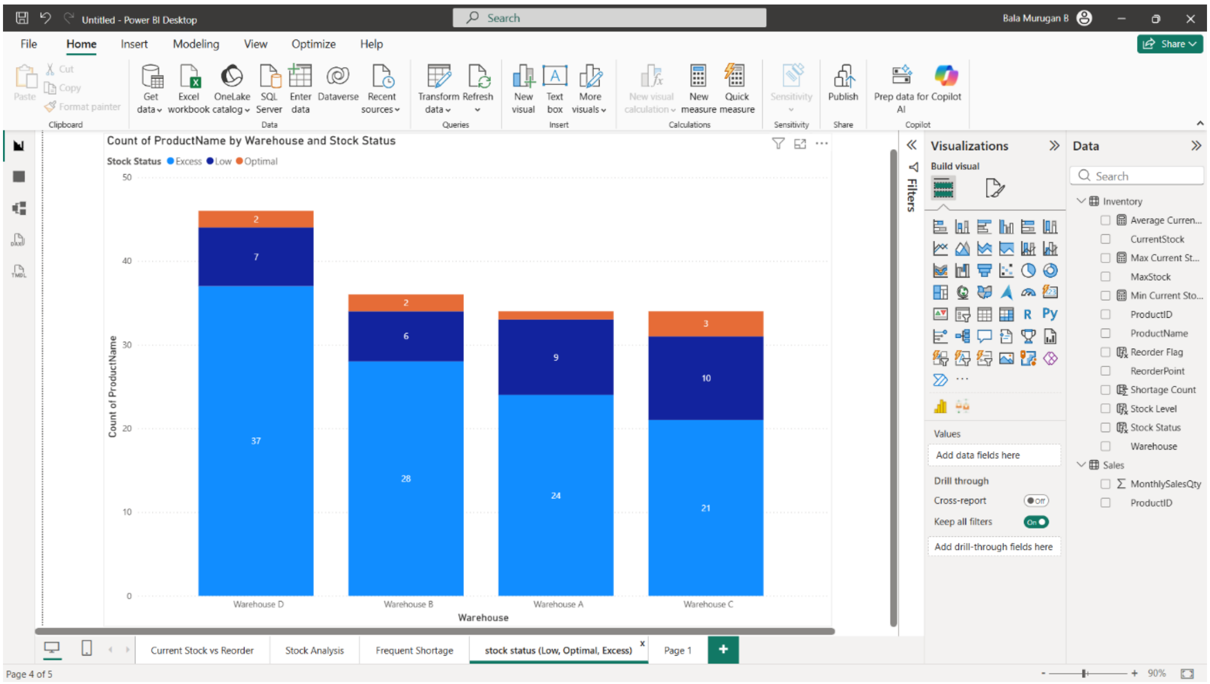
4.Create a DAX measure to classify stock status (Low, Optimal, Excess):

Low:Stock is below the Reorder Point, needs replenishment soon.

Optimal:Stock is between Reorder Point and Maximum Stock Level, ideal inventory level.

Excess:Stock exceeds Maximum Stock Level, leading to higher holding costs.

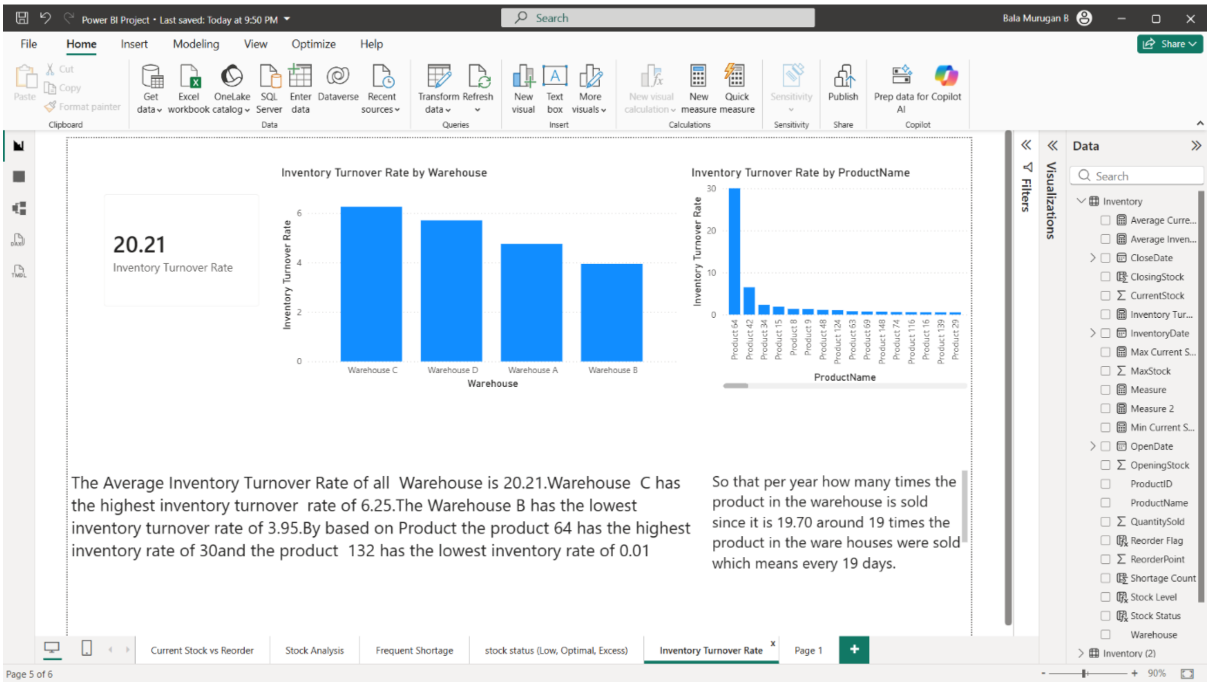




5.Calculate inventory turnover rate:

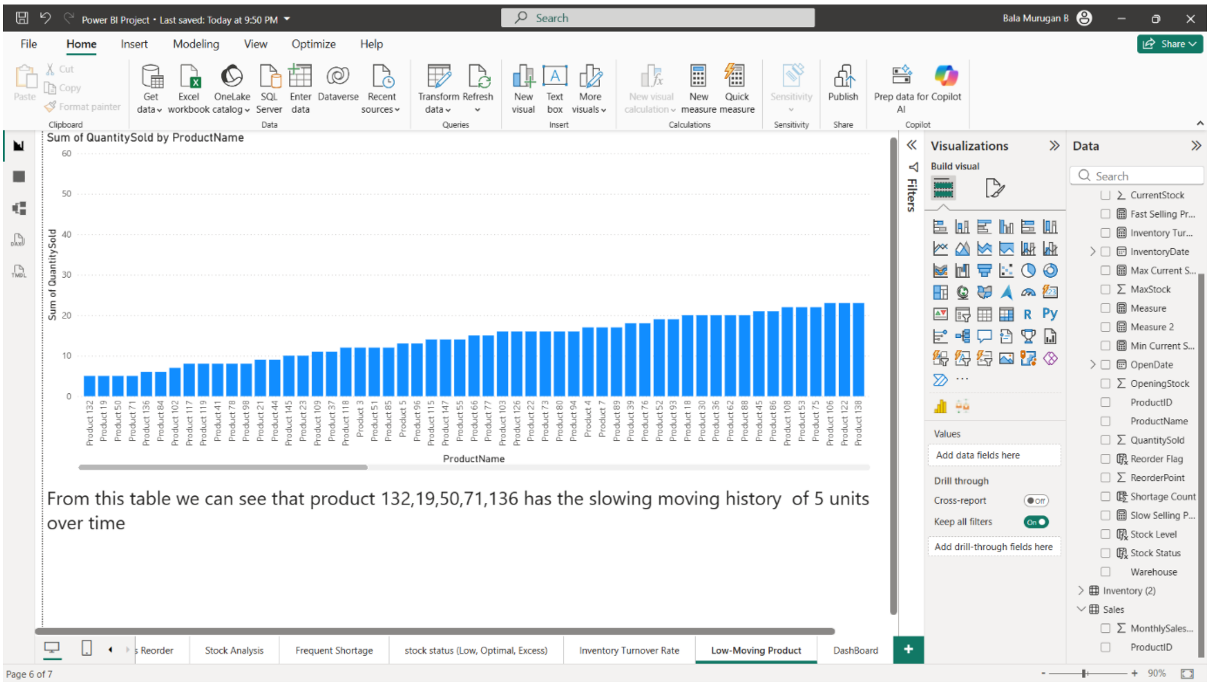
```
Inventory Turnover Rate =
DIVIDE(
    SUM(Inventory[QuantitySold]),
    [Average Inventory]
)
```

```
Average Inventory =
DIVIDE(
    AVERAGE(Inventory[OpeningStock]) +
    AVERAGE('Inventory'[ClosingStock]),
    2
)
```



6.Identify slow-moving products using sales and inventory data:

From this table we can see that product 132,19,50,71,136 has the slowing moving history of 5 units over time using the quantity sold column.



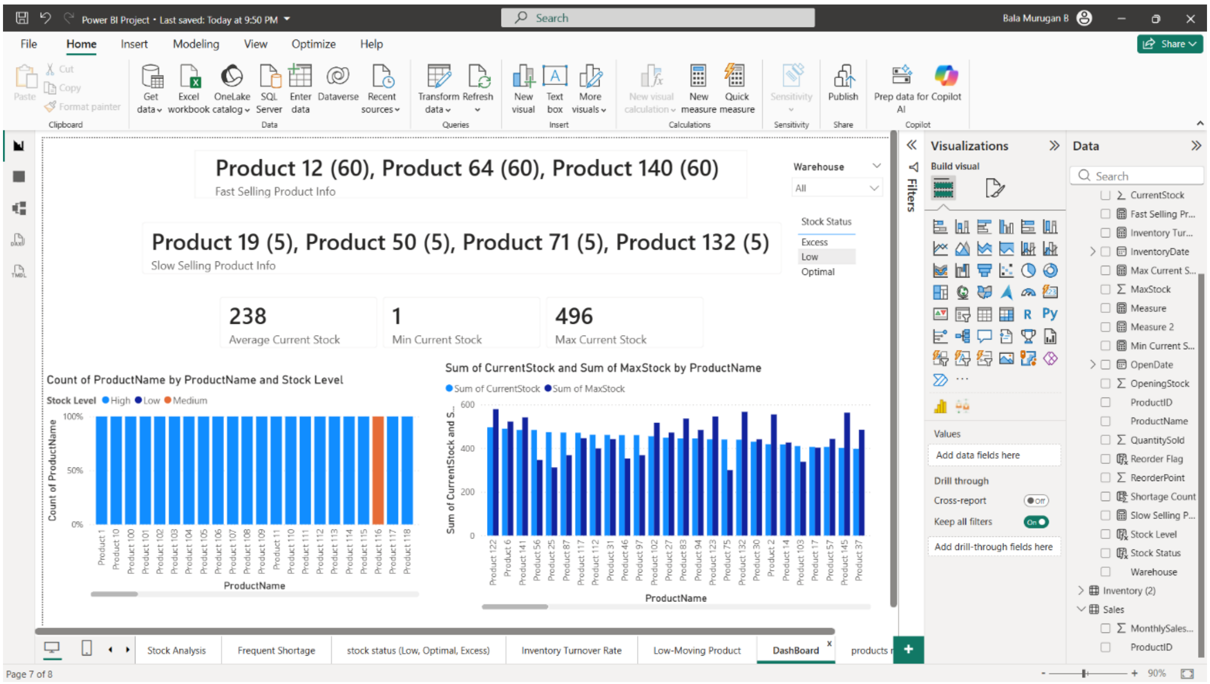
1. Design a dashboard to monitor stock health.

7.Design a dashboard to monitor stock health:

This dashboard provides a comprehensive overview of product inventory, highlighting fast and slow-selling products, stock levels, and overall inventory health. It combines key metrics, product insights, and visual analytics for quick decision-making.

Key Highlights at the Top

1. Fast Selling Products:
   - Shows product names with the highest quantity sold, along with the sold quantity in parentheses.
   - Example: Product 12 (60), Product 64 (60), Product 140 (60)
   - Helps identify high-demand products requiring regular replenishment.
2. Slow Selling Products:
   - Lists products with lowest quantity sold, along with the sold quantity.
   - Example: Product 19 (5), Product 50 (5), Product 71 (5), Product 132 (5)
   - Useful for inventory optimization and deciding discounts, promotions, or stock reduction.
3. Stock Status Filter:
   - Allows filtering products based on Low, Optimal, or Excess stock categories.



8.Which products require immediate replenishment?

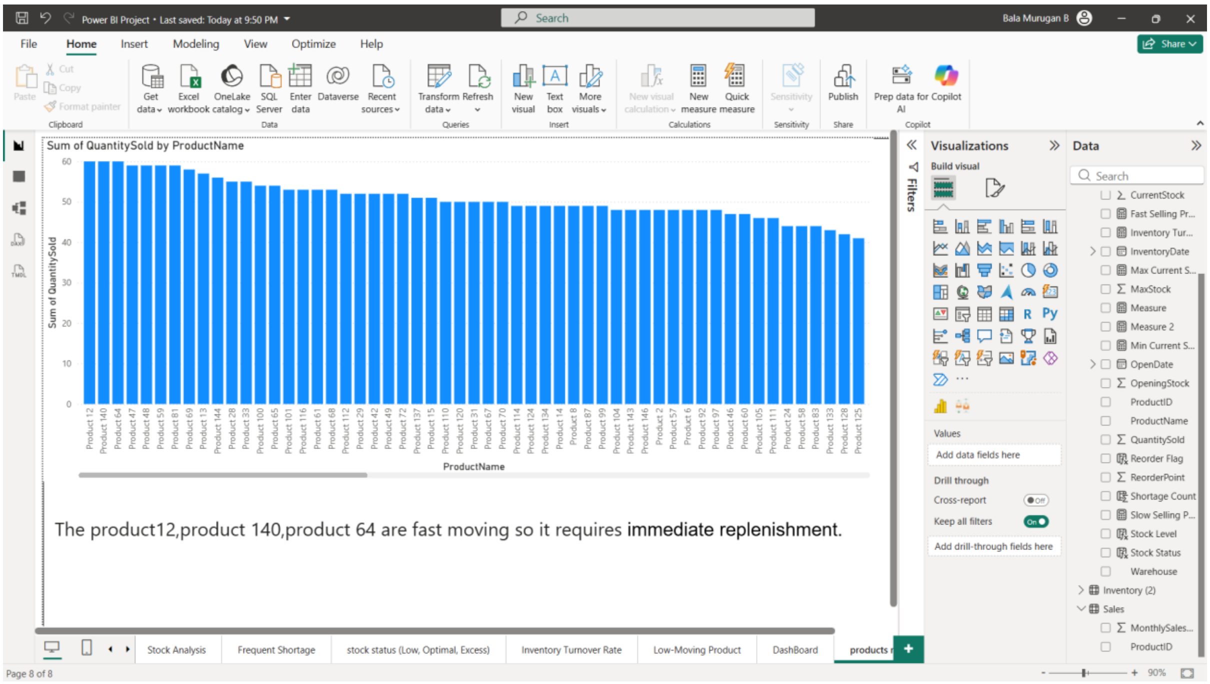
This report visualizes the sales performance of products and highlights fast-moving items that require immediate replenishment. It provides insights into which products are in high demand and helps prioritize inventory management decisions.

Bar Chart: Quantity Sold by Product

- X-axis: Product Name
- Y-axis: Sum of Quantity Sold
- Bars: Represent the total quantity sold for each product.

Insights from the chart:

- Products on the left side have the highest sales, indicating they are fast-moving products.
- Products on the right side have lower sales, representing slower-moving items.
- Example: Product 12, Product 140, Product 64 are the top-selling items.



9.How can inventory insights reduce holding costs?

Holding costs (or carrying costs) are the expenses a company incurs for storing inventory, including:

- Warehousing costs (rent, utilities)
- Depreciation or obsolescence of goods
- Insurance
- Capital tied up in inventory

Reducing excess inventory directly reduces these costs.

Inventory Level Monitoring

- **Use Power BI dashboards** to visualize current stock levels across warehouses.
- **Identify slow-moving or excess inventory** using charts like **bar charts or heat maps**.
- Example: A bar chart showing units in stock vs average monthly sales helps spot overstocked items.

Forecasting Demand

- Use **Power BI's forecasting capabilities** on historical sales data.
- Predict **future demand** and adjust purchase orders accordingly.
- This prevents overstocking items that tie up capital unnecessarily.

Identifying Overstock & Obsolete Items

- Use **Power BI filters and conditional formatting** to highlight items not sold for a long time.
- Highlight items with high stock but low sales velocity.
- Reducing these items lowers holding costs and prevents write-offs.

Optimizing Reorder Levels

- Power BI can calculate **economic order quantities (EOQ)** and **safety stock levels**.
- Dashboards can show items approaching reorder points, so you don't overstock.